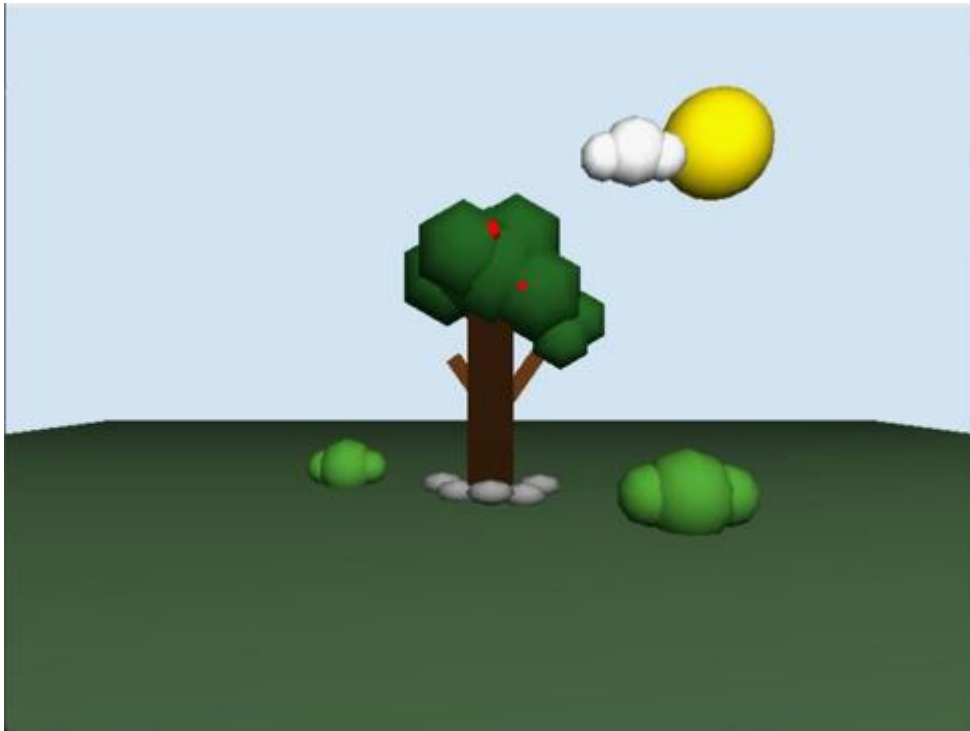




CS360 -Computer Graphics Lab Project
Second semester 1447- 2026

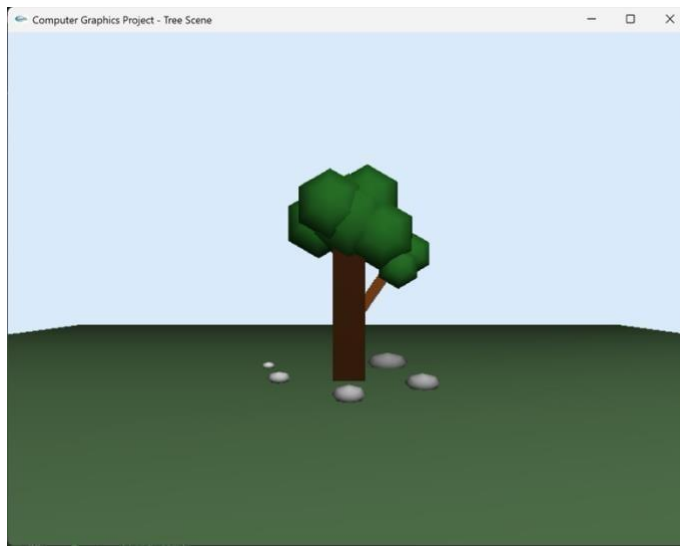


Project Group (wasn't assigned)



Name	ID
	
Hala Abdullatif Alghamdi	

Phase 1: Lighting, depth setup and first sketch



Design Description:

The design features a 3D natural scene centered around a stylized tree. It consists of a textured-looking trunk created using a scaled `glutSolidCube`, an angled side branch for realism, and a lush canopy formed by multiple overlapping `glutSolidSphere` objects. To add environmental detail, the scene includes a flat green terrain (ground plane) and several scattered stones of varying sizes at the base of the tree. The scene is interactive, allowing the user to rotate the view or zoom in/out to explore the 3D space from different perspectives.

Light and Depth Setup:

To achieve a sense of volume and spatial order, we enabled `GL_DEPTH_TEST`, which ensures that closer objects correctly obscure those further away. For lighting, we implemented a positional light source using `GL_LIGHT0`, placed at specific coordinates to create highlights and shadows on the tree's geometry. We also utilized `GL_COLOR_MATERIAL` to ensure that the assigned colors of the objects interact naturally with the light source, giving the 3D shapes a more solid and vibrant appearance.

Challenges and Solutions:

The main challenge was the spatial coordination of the tree's components. Specifically, determining the exact 3D coordinates for the trunk and the multiple spheres of the canopy was difficult, as it was hard to visualize how they would align from every angle. Initially, the leaves appeared disconnected or floating away from the trunk. We solved this by using a trial-and-error approach with the `glTranslatef` and `glPushMatrix` functions, incrementally adjusting the X, Y, and Z positions. We also utilized the interactive camera we built to rotate around the model during development, which helped us ensure that the branches and leaves were perfectly centered and looked natural from all perspectives.

Phase 2: Model Construction



3D Primitive Modeling Description :

Our model is constructed using a hierarchical assembly of standard 3D primitives to create a cohesive natural scene. The core of the tree is built using a primary `glutSolidCube` scaled into a tall rectangular prism for the trunk, complemented by two additional cubes that are rotated and scaled to form the main structural branches. The canopy is designed using a series of overlapping `glutSolidSphere` objects of varying radii, creating a lush, organic volume. Beyond the tree, we utilized spheres to model complex environmental elements: clouds and grass tufts are formed by grouping multiple spheres together, while the sun and apples are represented by individual high-precision spheres. Finally, the terrain is established using a large `GL_QUADS` plane, and the rocks are created using spheres that are non-uniformly scaled via `glScalef` to achieve a flattened, realistic appearance.

Challenges and Solutions:

The primary challenge encountered during this phase was the precise spatial coordination and alignment of the tree's various components within the 3D coordinate system. Specifically, it was difficult to determine the exact X, Y, and Z translations for the apples and branches so that they would appear naturally attached to the trunk and nested within the canopy without "floating" in mid-air. Initially, parts of the tree looked disconnected or incorrectly layered when rendering the scene.

We addressed this through a systematic "trial-and-error" approach using the `glTranslatef` and `glPushMatrix/glPopMatrix` functions. By making incremental adjustments to the coordinate values and carefully managing the transformation stack, we were able to precisely place each element. This iterative process of refining the mathematical offsets ensured that the branches, leaves, and fruits formed a singular, balanced structure where every primitive is correctly positioned in relation to the others.

Phase 3: Basic Animation & Interaction



3D Primitive Modeling:

The scene is constructed using a hierarchical assembly of basic shapes. The tree trunk and its branches are built from scaled and rotated `glutSolidCube` primitives. The canopy, apples, clouds, and grass are created by grouping multiple `glutSolidSphere` objects of varying sizes. To enhance the environment, we used a large `GL_QUADS` for the ground and non-uniformly scaled spheres (`glScalef`) to create flattened river rocks.

Animation & Interaction:

- Animation: We implemented a continuous horizontal "drifting" effect for the clouds that reverses direction upon reaching a boundary. We also added a vertical falling animation for the apple to simulate gravity.
- Keyboard Interaction: We added the 'F' key functionality to trigger the apple's fall. The 'Space' key remains the primary method to reset the scene's parameters to their default state.

Design Improvements:

The environment was polished by balancing the sky-blue background with a vibrant green terrain. By adding compound objects like "fluffy" clouds and scattered rocks at the tree's base, we moved beyond a simple model to create a more detailed and natural 3D atmosphere.

Challenges & Solutions:

The main challenge was spatial coordination. It was difficult to calculate the exact X, Y, and Z offsets to ensure the branches and apples were perfectly attached to the trunk. We solved this through a systematic trial-and-error approach, incrementally adjusting the translation values until all primitives were seamlessly aligned and the scene felt structurally sound.

Final Reflection

During this project, our team gained a deeper understanding of the hierarchical structure of 3D modeling and the complexities of coordinating multiple primitives within a shared coordinate system. The most challenging part was achieving precise spatial alignment for the tree's components, while the most rewarding aspect was successfully implementing the animation and the 'F' key interaction, seeing our static model come to life. If we had more time, we would improve the scene by adding more complex textures to the trunk and terrain, and we would expand the environment with additional animated elements like falling leaves or a more detailed particle system for the clouds.

Work distribution table

	Constructed the core structural components of the tree (Trunk and main branches) using scaled cube primitives. Developed the horizontal movement logic for the clouds to create a continuous drifting effect. Set up the scene's environment including the sky background color and the green terrain plane. Modeled the ground details by designing the non-uniformly scaled river rocks.
Hala Abdullatif Alghamdi	Implemented the lighting system and material properties to ensure realistic 3D volume and shading. Designed the organic elements of the model, including the multi-layered leaf canopy and compound grass structures. Programmed the interactive apple-fall animation and handled the 'F' key event within the keyboard function. Modeled decorative celestial and environmental objects, specifically the sun, the apples, and the fluffy cloud clusters.
Joint Effort (Both)	Performed iterative spatial coordination to align primitives and ensure no floating objects. Conducted debugging for transformation stack issues (glPushMatrix / glPopMatrix). Finalized the 3D scene aesthetics and documented the project challenges and solutions.